

Week 1: Python Basics

Input, Output, Variables, Logic & If/Else

Bit by Bit Coding

1 Output and Variables

Programs need to store data and communicate with the user.

💡 The Print Function

The `print()` function displays text or numbers on the screen.

🔗 Output Examples

```
1 print("Hello World")
2 print("HP:", 100)      # Commas insert a space automatically
3 print("Key " + "Found") # Plus sign joins strings together
```

💡 Variables

A variable is a **named box** that stores data. You can change what's inside the box at any time.

Naming Rules:

- Use underscores for spaces (e.g., `player_name`)
- Cannot start with a number
- Case-sensitive (`Score` $\neq$ `score`)

🔗 Using Variables

```
1 name = "Zelda"      # Create a variable
2 print(name)         # Use it
3
4 name = "Link"       # Overwrite it with new data
5 print(name)         # Now prints "Link"
```

2 Data Types and Input

Every value in Python has a specific type.

- **String (str):** Text (e.g., `"Hello"`, `'123'`)
- **Integer (int):** Whole numbers (e.g., `42`, `-5`)
- **Float:** Decimal numbers (e.g., `3.14`, `0.5`)

- **Boolean (bool):** Logical truth (e.g., True, False)

⚠ Common Mistake: Strings vs Numbers

"10" + "5" results in "105" (text sticking together).
10 + 5 results in 15 (math).

💡 Input Function

The `input()` function pauses the program and waits for the user to type something. **It always returns a string.**

🔗 Getting User Input

```
1 # Get text
2 name = input("Enter your name: ")
3
4 # Get a number (must convert it!)
5 age = int(input("Enter your age: "))
```

3 Operators

Operators allow you to perform math and comparisons.

💡 Arithmetic Operators

- + (Add), - (Subtract), * (Multiply), / (Divide)
- // (Floor Division: rounds down to nearest whole number)
- % (Modulus: gives the remainder)
- ** (Exponent: power of)

🔗 Arithmetic Examples

```
1 health = 100
2 damage = 15
3
4 health = health - damage # Update variable
5 # Shorthand:
6 health -= damage        # Same as above!
7
8 print(10 // 3) # Prints 3
9 print(10 % 3)  # Prints 1
```

💡 Comparison Operators

These always return a **Boolean** (True or False).

- == (Equal to), != (Not equal to)
- > (Greater), < (Less), >=, <=

⚠ Assignment vs Comparison

= assigns a value to a variable.
== compares two values.

4 Logic and Control Flow

Control flow lets your code make decisions.

💡 Logical Operators

- **and**: Both sides must be True.
- **or**: At least one side must be True.
- **not**: Flips True to False (and vice versa).

🐍 Logic Example

```
1 has_key = True
2 has_torch = False
3
4 # Evaluates to False because one is missing
5 can_enter = has_key and has_torch
```

💡 If / Elif / Else Statements

Use indentation (4 spaces) to define blocks of code that run only if a condition is true.

🐍 Making Decisions

```
1 score = 85
2
3 if score >= 90:
4     print("Grade: A")
5 elif score >= 80:
6     print("Grade: B")
7 else:
8     print("Grade: C")
```